National University of Computer & Emerging Sciences, KarachiComputer Science Department
Spring 2024, Lab Manual – 11

Course Code: CL-2005	Course : Database Systems Lab
Instructor(s) :	Fatima Gado

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Overview of MongoDB

MongoDB is an open-source document database and leading NoSQL database. MongoDB is written in C++. This manual will give you great understanding on MongoDB concepts needed to create and deploy a highly scalable and performance-oriented database.

MongoDB is a cross-platform, document oriented database that provides, high performance, high availability, and easy scalability. MongoDB works on concept of collection and document.

Difference in Terminology of MongoDB

RDBMS	MongoDB
Database	Database
Table	Collection
Tuple/Row	Document
column	Field
Table Join	Embedded Documents
Primary Key	Primary Key (Default key _id provided by mongodb itself)
Database Server and Client	
Mysqld/Oracle	mongod
mysql/sqlplus	mongo

Figure 1(Difference between RDBMS & MongoDB)

Database

Database is a physical container for collections. Each database gets its own set of files on the file system. A single MongoDB server typically has multiple databases.

Collection

Collection is a group of MongoDB documents. It is the equivalent of an RDBMS table. A collection exists within a single database. Collections do not enforce a schema. Documents within a collection can have different fields. Typically, all documents in a collection are ofsimilar or related purpose.

Document

A document is a set of key-value pairs. Documents have dynamic schema. Dynamic schemameans that documents in the same collection do not need to have the same set of fields or structure, and common fields in a collection's documents may hold different types of data.

Install MongoDB on Windows

To perform the installation of MongoDB do following the below steps:

1. Install MongoDB from official website.

   https://www.mongodb.com/try/download/community

2. Install MongoDB Shell from official website

   https://www.mongodb.com/try/download/shell

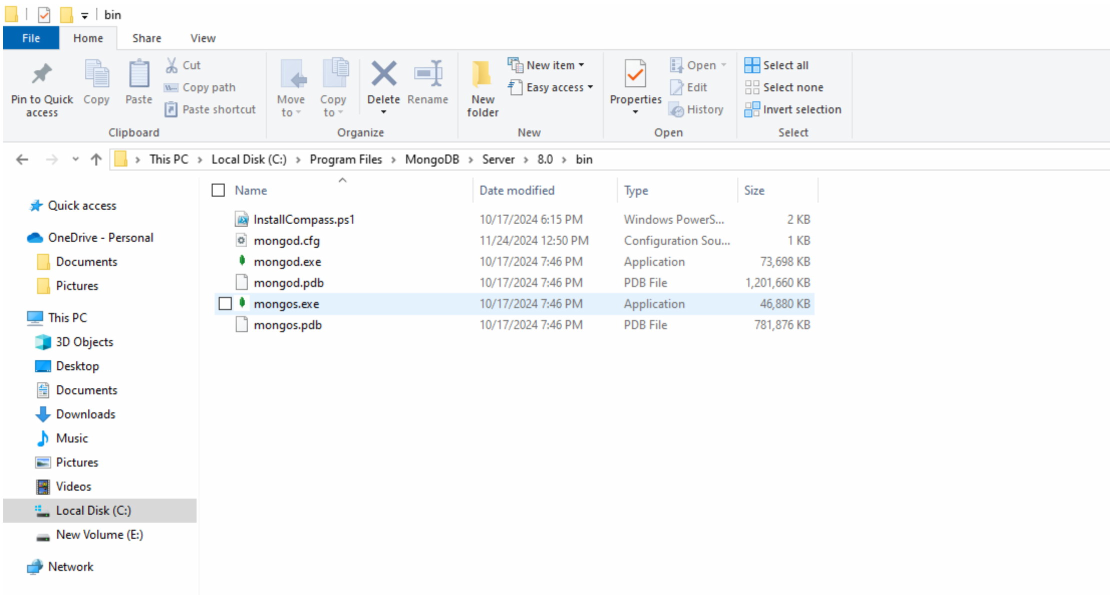
   Note: (Download both in .msi format).

   Install Mongo shell create a folder in c drive with name “mongosh” (Don’t use by default path given during the installation).

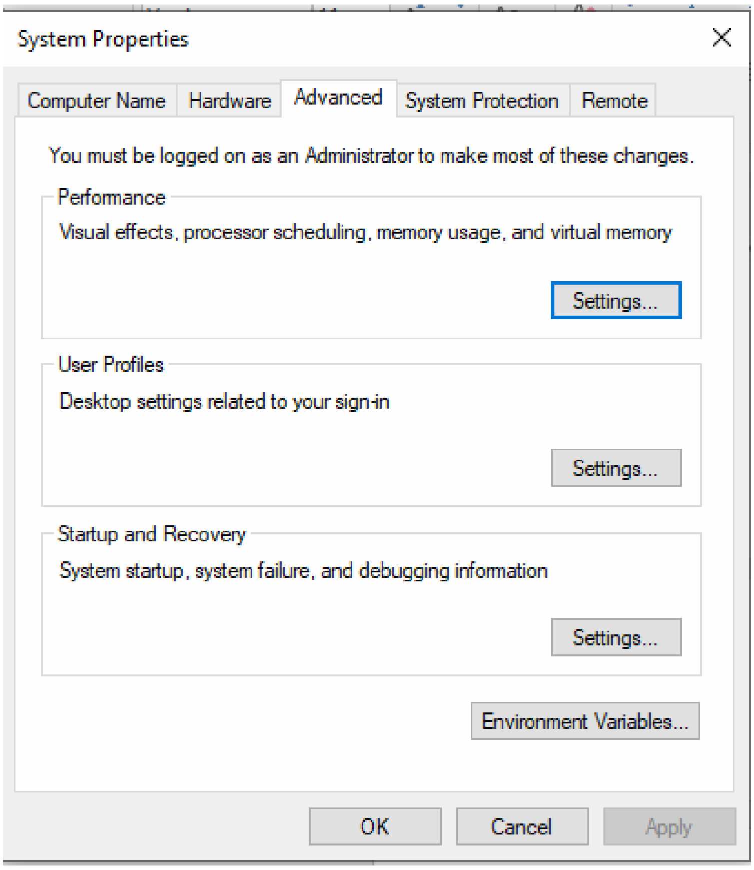
3. Now set path for both softwares in environmental variables.

   For MongoDB Copy the path where it is located

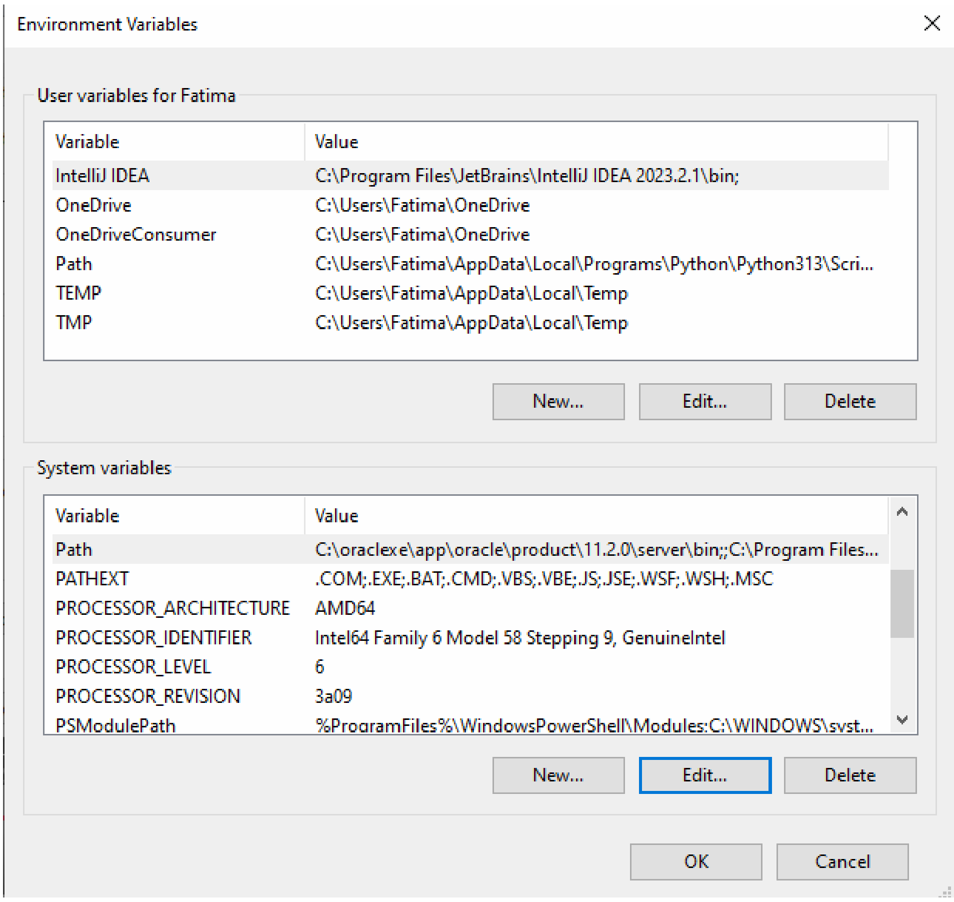
   C:\Program Files\MongoDB\Server\8.0\bin



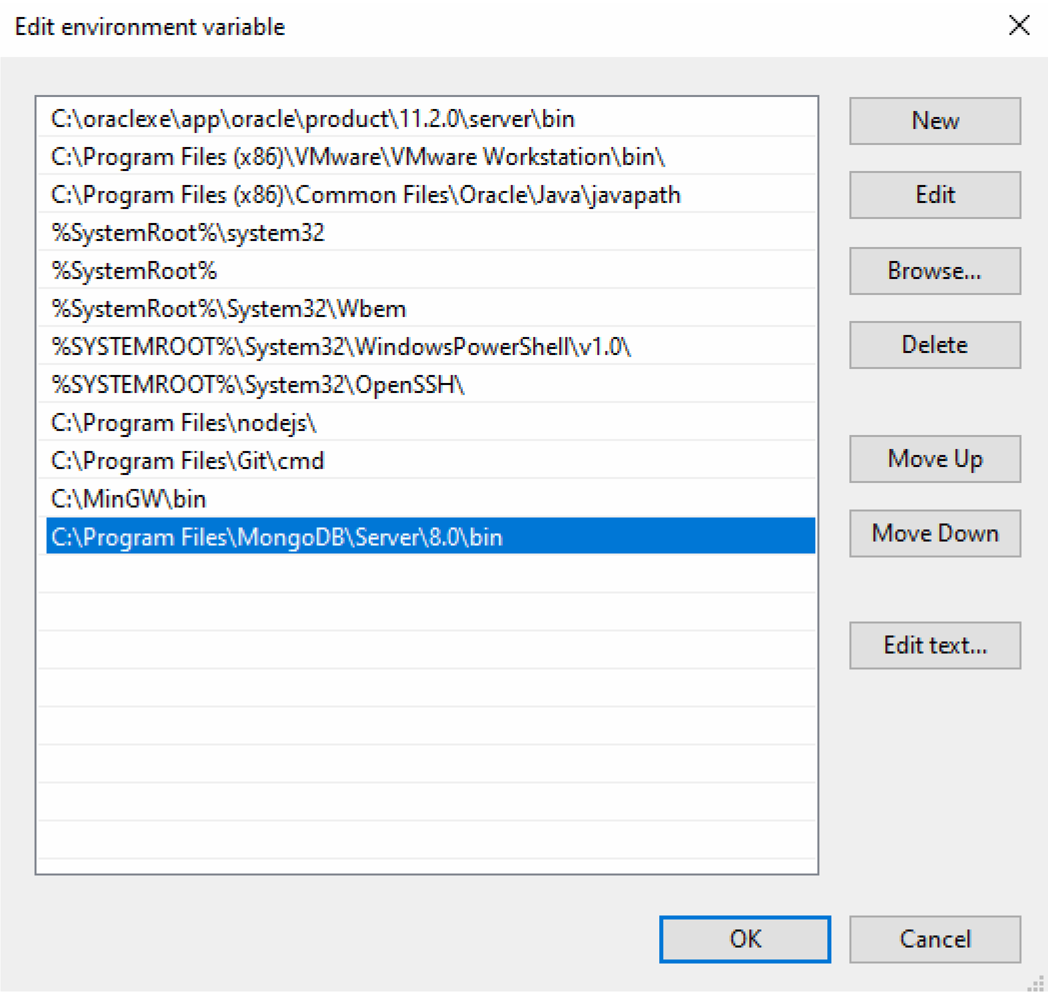
4. Search edit the system environment for setting path
5. Then go to the environmental variables



Select path from the System Variable and press edit



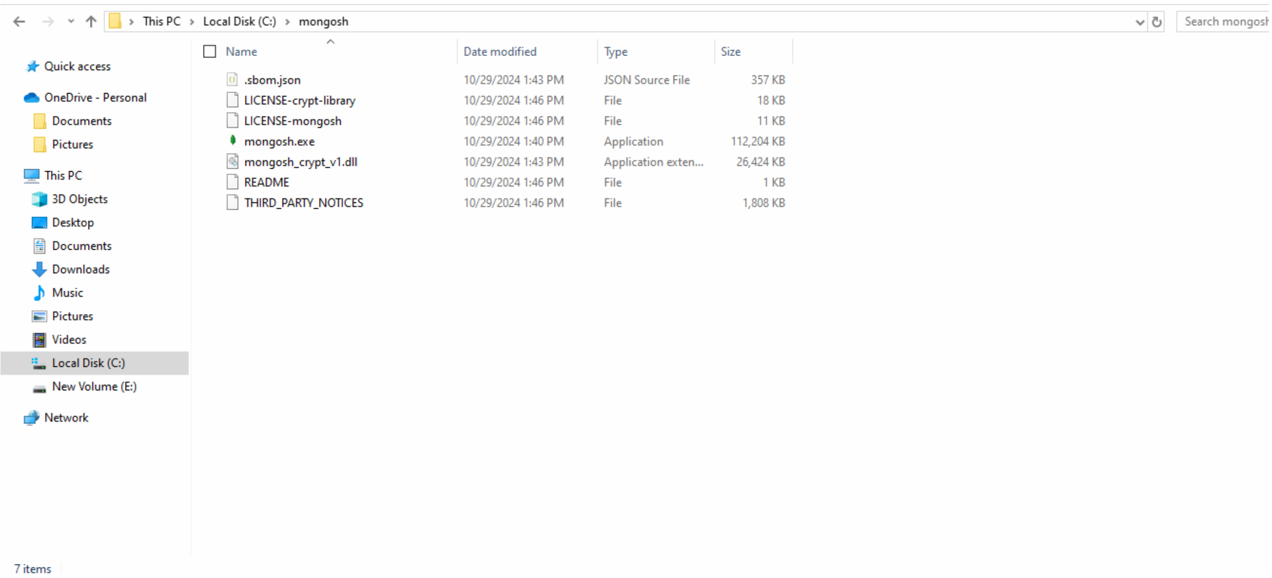
Click new and paste the path of mongoDB



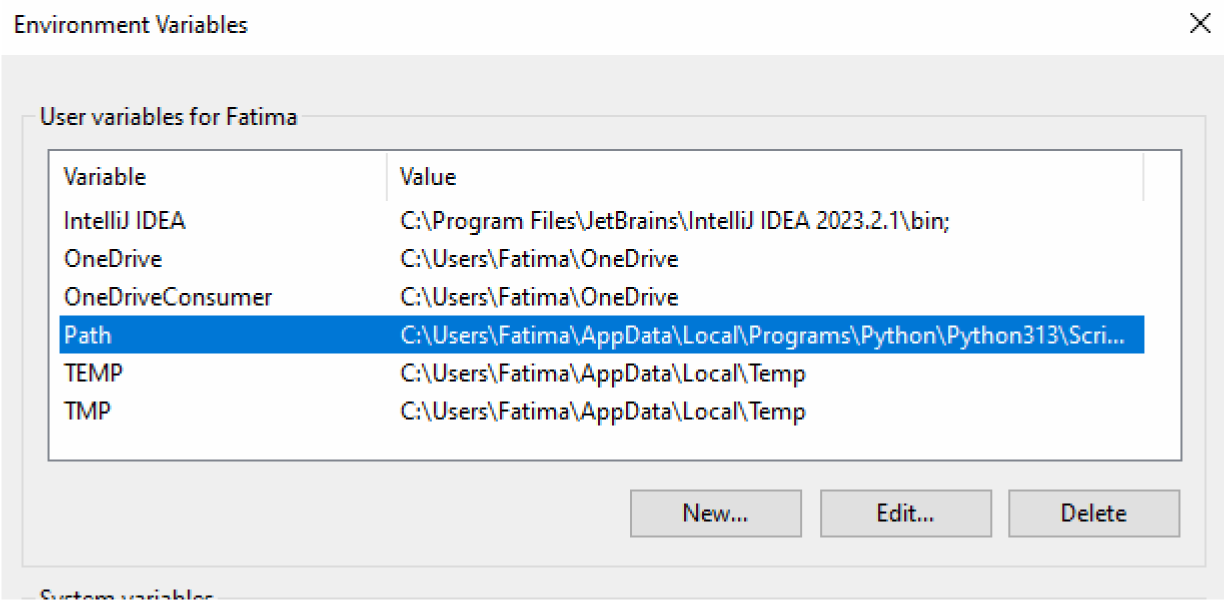
Then press OK.

6. Now repeat same for mongo shell.

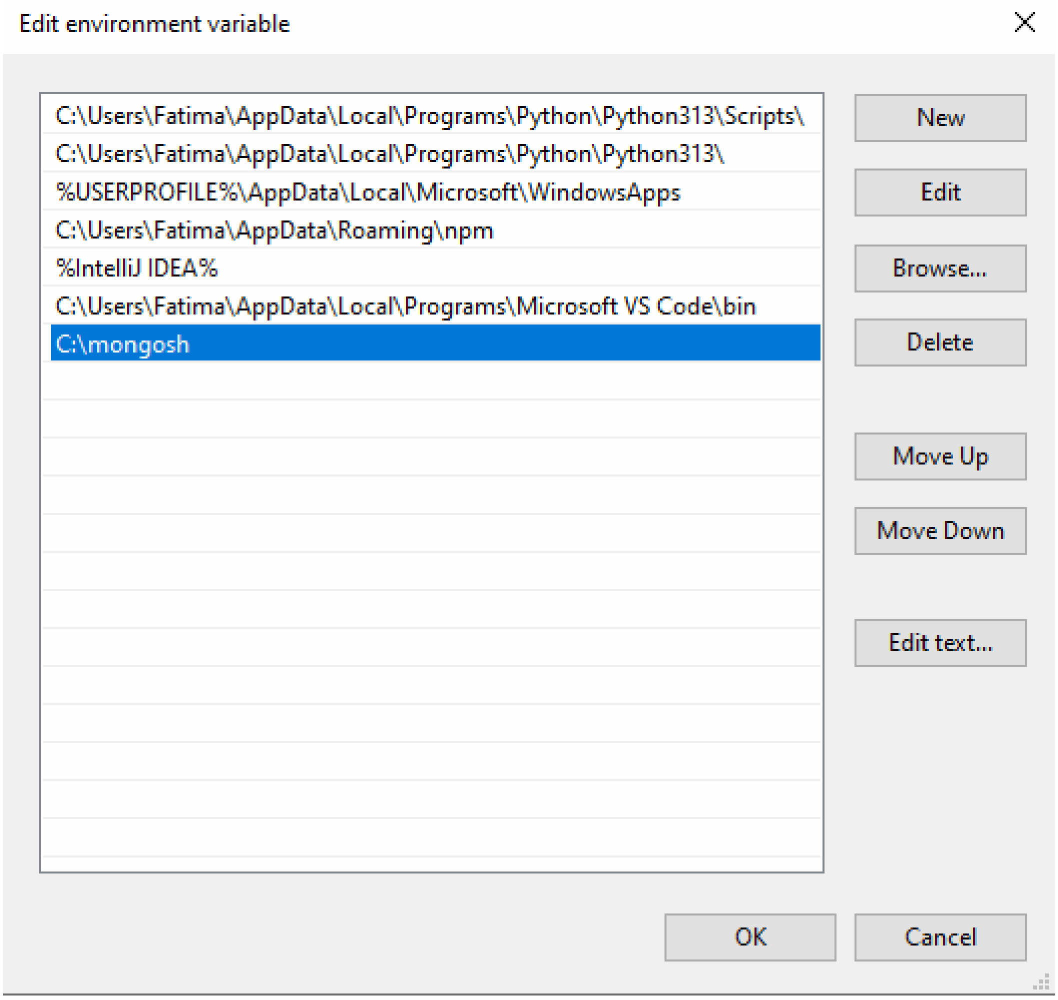
Copy path " C:\mongosh



Now select path from User Variable and press Edit



Now press New and paste path of mongo Shell



Now press Ok.

7. Open cmd an type mongosh

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Microsoft Windows [Version 10.0.19045.5011]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Fatima>mongosh
Current Mongosh Log ID: 6742e6a9765119dac60d818f
Connecting to:          mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2
.3.3
Using MongoDB:          8.0.3
Using Mongosh:          2.3.3

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/


To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.co
m/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

------
   The server generated these startup warnings when booting
   2024-11-24T12:50:34.638+05:00: Access control is not enabled for the database. Read and write access to data and conf
iguration is unrestricted
------

test>
```

8. After Successfully installation and path setting we use mongoshell and test it.

   For test type

   Show dbs

```
test> show dbs
admin   40.00 KiB
config  12.00 KiB
local   40.00 KiB
test>
```

9. Now running **db.help()** to see different helping functions

```
Command Prompt - mongo
> db.help()
DB methods:
        db.adminCommand(nameOrDocument) - switches to 'admin' db, and runs command [ just calls db.runCommand(...) ]
        db.auth(username, password)
        db.cloneDatabase(fromhost)
        db.commandHelp(name) returns the help for the command
        db.copyDatabase(fromdb, todb, fromhost)
        db.createCollection(name, { size : ..., capped : ..., max : ... } )
        db.createView(name, viewOn, [ { $operator: {...}}, ... ], { viewOptions } )
        db.createUser(userDocument)
        db.currentOp() displays currently executing operations in the db
        db.dropDatabase()
        db.eval() - deprecated
        db.fsyncLock() flush data to disk and lock server for backups
        db.fsyncUnlock() unlocks server following a db.fsyncLock()
        db.getCollection(cname) same as db['cname'] or db.cname
        db.getCollectionInfos([filter]) - returns a list that contains the names and options of the db's collections
        db.getCollectionNames()
        db.getLastError() - just returns the err msg string
        db.getLastErrorObj() - return full status object
        db.getLogComponents()
        db.getMongo() get the server connection object
        db.getMongo().setSlaveOk() allow queries on a replication slave server
        db.getName()
        db.getPrevError()
        db.getProfilingLevel() - deprecated
        db.getProfilingStatus() - returns if profiling is on and slow threshold
        db.getReplicationInfo()
        db.getSiblingDB(name) get the db at the same server as this one
        db.getWriteConcern() - returns the write concern used for any operations on this db, inherited from server object if set
        db.hostInfo() get details about the server's host
        db.isMaster() check replica primary status
        db.killOp(opid) kills the current operation in the db
        db.listCommands() lists all the db commands
        db.loadServerScripts() loads all the scripts in db.system.js
        db.logout()
```

Command Prompt - mongo

```
> db.stats()
{
        "db" : "test",
        "collections" : 1,
        "views" : 0,
        "objects" : 1,
        "avgObjSize" : 33,
        "dataSize" : 33,
        "storageSize" : 16384,
        "numExtents" : 0,
        "indexes" : 1,
        "indexSize" : 16384,
        "ok" : 1
}
>
```

10.Now run **db.stats()** to see the statistics of the database

Some Consideration While Designing Schemain MongoDB

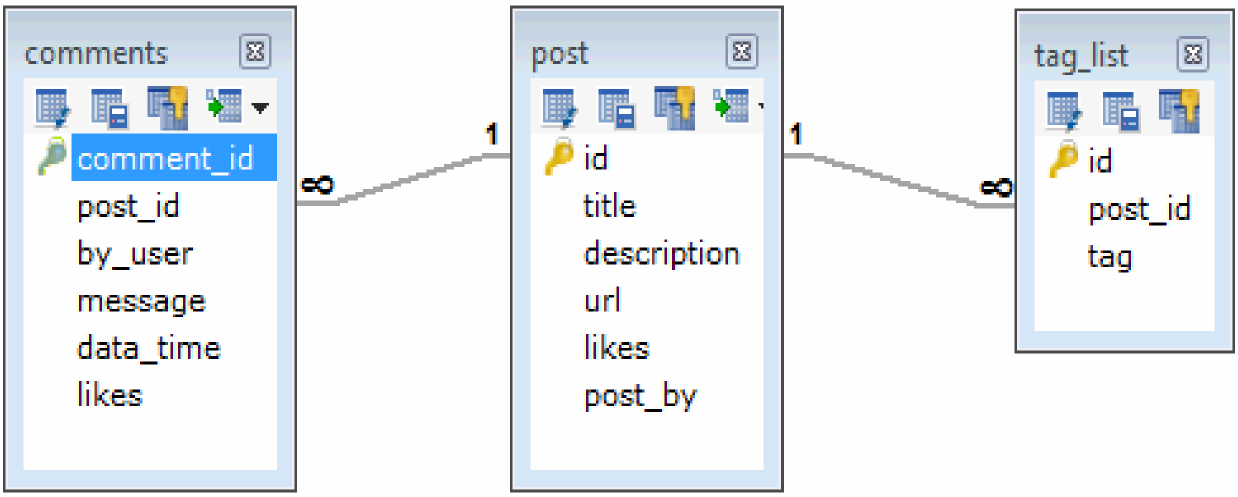
1. Design your schema according to user requirements.
2. Combine objects into one document if you will use them together. Otherwise separate them (but make sure there should not be need of joins).
3. Duplicate the data (but limited) because disk space is cheap as compare to compute time.
4. Do joins while write, not on read.
5. Optimize your schema for most frequent use cases.
6. Do complex aggregation in the schema

Example

Suppose a client needs a database design for his blog/website and see the differences between RDBMS and MongoDB schema design. Website has the following requirements.

1. Every post has the unique title, description and url.
2. Every post can have one or more tags.
3. Every post has the name of its publisher and total number of likes.
4. Every post has comments given by users along with their name, message, data-time andlikes.
5. On each post, there can be zero or more comments

In RDBMS schema, design for above requirements will have minimum three tables.



While in MongoDB schema, design will have one collection post and the following structure:

```
{
   _id: POST_ID
   title: TITLE_OF_POST,
   description: POST_DESCRIPTION,
   by: POST_BY,
   url: URL_OF_POST,
   tags: [TAG1, TAG2, TAG3],
   likes: TOTAL_LIKES,
   comments: [
      {
         user:'COMMENT_BY',
         message: TEXT,
         dateCreated: DATE_TIME,
         like: LIKES
      },
      {
         user:'COMMENT_BY',
         message: TEXT,
         dateCreated: DATE_TIME,
         like: LIKES
      }
   ]
}
```

So while showing the data, in RDBMS you need to join three tables and in MongoDB, data will be shown from one collection only.

MongoDB – Create Database

MongoDB **use DATABASE_NAME** is used to create database. The command will create a new database if it doesn't exist, otherwise it will return the existing database.

Syntax

Basic syntax of **use DATABASE** statement is as follows – use DATABASE_NAME

```
Command Prompt - mongo
> use myfirstdb
switched to db myfirstdb
>
```

Command:

>show dbs

If you want to check your databases list, use the command **show dbs**.

```
Command Prompt - mongo
> use myfirstdb
switched to db myfirstdb
> show dbs
admin  0.000GB
local  0.000GB
test   0.000GB
>
```

Your created database (myfirstdb) is not present in list. To display database, you need to insert at least one document into it.

> db.myfirstdb.insert({"name":"Amin Sadiq"})WriteResult({ "nInserted" : 1 })

The dropDatabase() Method

MongoDB db.dropDatabase() command is used to drop an existing database

Syntax

Basic syntax of dropDatabase() command is as follows –

> db.dropDatabase()

This will delete the selected database. If you have not selected any database, then it will delete default 'test' database.

Example

If you want to delete new database <mydb>, then dropDatabase() command would be as follows –

>use myfirstdb

// Switched to db mydb

>db.dropDatabase()

{ "dropped" : "myfirstdb", "ok" : 1 }

Now check list of databases.

>show dbs

The createCollection() Method

MongoDB db.createCollection(name, options) is used to create collection.

Syntax

Basic syntax of createCollection() command is as follows –

db.createCollection(name, options)

In the command, name is name of collection to be created. Options (Optional Parameter)isa document and is used to specify configuration of collection.

```
Command Prompt - mongo
> use myfirstdb
switched to db myfirstdb
> show dbs
admin  0.000GB
local  0.000GB
test   0.000GB
> db.movie.insert({"name":"Amin Sadiq"})
WriteResult({ "nInserted" : 1 })
> show dbs
admin      0.000GB
local      0.000GB
myfirstdb  0.000GB
test       0.000GB
> db.createcollection("myCollection")
2021-11-20T12:16:51.393+0500 E QUERY    [thread1] TypeError: db.createcollection is not a function :
@(shell):1:1
> db.createCollection("myCollection")
{ "ok" : 1 }
> show collections
movie
myCollection
>
```

MongoDB – Drop Collection

The drop() Method

MongoDB's **db.collection.drop()** is used to drop a collection from the database

Syntax

Basic syntax of **drop()** command is as follows –

db.COLLECTION_NAME.drop()

```
> show collections
movie
myCollection
> db.movie.drop()
true
> show dbs
admin      0.000GB
local      0.000GB
myfirstdb  0.000GB
test       0.000GB
> show collections
myCollection
>
```

MongoDB – Insert Document

The insert() Method

To insert data into MongoDB collection, you need to use MongoDB's insert() or save() method

Syntax

The basic syntax of insert() command is as follows –

>db.COLLECTION_NAME.insert(document)

Example

```
> db.mycollection.insert({
    title: 'MongoDB
    Overview',
    description: 'MongoDB is no sql database',
    by: 'Amin Sadiq',
    url: 'http://www.gmail.com',
    tags: ['mongodb', 'database', 'NoSQL'],
    likes: 100
    })
```

Notepad: Here **mycollection** is our collection name, as created in the previous slide. If the collection doesn't exist in the database, then MongoDB will create this collection and theninsert a document into it.

_id parameter
In the inserted document, if we don't specify the _id parameter, then MongoDB assigns a unique ObjectId for this document.

Insert Document multiple document

To insert multiple documents in a single query, you can pass an array of documents in insert()command.

```
db.mycollection.insert([
  {
    title: 'MongoDB Overview',
    description: 'MongoDB is no sql database',
    by: 'Amin Sadiq',
    url: 'http://www.gmail.com',
    tags: ['mongodb', 'database', 'NoSQL'],
    likes: 100
  },

  {
    title: 'NoSQL Database',
    description: "NoSQL database doesn't have tables",
    by: 'Ali Shah Fatmi',
    url:  'http://www.gmail.com',
    tags: ['mongodb', 'database', 'NoSQL'],
    likes: 20,
    comments: [
      {
        user:'user1',
        message: 'My first comment',
        dateCreated: new
        Date(2021,11,10,2,35),like: 0
      }
    ]
  }
])
```

```
Command Prompt - mongo
> db.mycollection.insert([
...    {
...       title: 'MongoDB Overview',
... description: 'MongoDB is no sql database',
... by: 'Amin Sadiq',
... url: 'http://www.gmail.com',
... tags: ['mongodb', 'database', 'NoSQL'],
... likes: 100
...    },
...
...    {
...       title: 'NoSQL Database',
...       description: "NoSQL database doesn't have tables",
...       by: 'Ali Shah Fatmi',
...       url: 'http://www.gmail.com',
...       tags: ['mongodb', 'database', 'NoSQL'],
...       likes: 20,
...       comments: [
...          {
...             user:'user1',
...             message: 'My first comment',
...             dateCreated: new Date(2021,11,10,2,35),
...             like: 0
...          }
...       ]
...    }
... ])
BulkWriteResult({
        "writeErrors" : [ ],
        "writeConcernErrors" : [ ],
        "nInserted" : 2,
        "nUpserted" : 0,
        "nMatched" : 0,
        "nModified" : 0,
        "nRemoved" : 0,
        "upserted" : [ ]
})
>
```

MongoDB – Query Document

The find() Method

To query data from MongoDB collection, you need to use MongoDB's find() method.

Syntax

- The basic syntax of find() method is as follows –
- >db.COLLECTION_NAME.find()

```
Command Prompt - mongo
> db.mycollection.find()
{ "_id" : ObjectId("6198f3f41388304875b893e7"), "title" : "MongoDB Overview", "description" : "MongoDB is no sql database", "by" : "Amin Sadiq", "url" : "http://www.gmail.com", "tags" : [ "mongodb", "database", "NoSQL" ], "likes" : 100 }
{ "_id" : ObjectId("6198f3f41388304875b893e8"), "title" : "NoSQL Database", "description" : "NoSQL database doesn't have tables", "by" : "Ali Shah Fatmi", "url" : "http://www.gmail.com", "tags" : [ "mongodb", "database", "NoSQL" ], "likes" : 20, "comments" : [ { "user" : "user1", "message" : "My first comment", "dateCreated" : ISODate("2021-12-09T21:35:00Z"), "like" : 0 } ] }
>
```

Note: **find()** method will display all the documents in a non-structured way.

The pretty() Method

To display the results in a formatted way, you can use pretty() method.

Syntax

- >db.mycol.find().pretty()

```
Command Prompt - mongo
> db.mycollection.find().pretty()
{
        "_id" : ObjectId("6198f3f41388304875b893e7"),
        "title" : "MongoDB Overview",
        "description" : "MongoDB is no sql database",
        "by" : "Amin Sadiq",
        "url" : "http://www.gmail.com",
        "tags" : [
                "mongodb",
                "database",
                "NoSQL"
        ],
        "likes" : 100
}
{
        "_id" : ObjectId("6198f3f41388304875b893e8"),
        "title" : "NoSQL Database",
        "description" : "NoSQL database doesn't have tables",
        "by" : "Ali Shah Fatmi",
        "url" : "http://www.gmail.com",
        "tags" : [
                "mongodb",
                "database",
                "NoSQL"
        ],
        "likes" : 20,
        "comments" : [
                {
                        "user" : "user1",
                        "message" : "My first comment",
                        "dateCreated" : ISODate("2021-12-09T21:35:00Z"),
                        "like" : 0
                }
        ]
}
>
```

RDBMS Where Clause Equivalents in MongoDB

Operation	Syntax	Example	RDBMS Equivalent
Equality	{<key>:<value>}	db.mycollection.find({"by":"Amin Sadiq"}).pretty()	where by = 'Amin Sadiq'
Less Than	{<key>:{$lt:<value>}}	db.mycollection.find({"likes":{$lt:50}}).pretty()	where likes < 50
Less Than Equals	{<key>:{$lte:<value>}}	db.mycollection.find({"likes":{$lte:50}}).pre tty()	where likes <= 50
Greater Than	{<key>:{$gt:<value>}}	db.mycollection.find({"likes":{$gt:50}}).pret ty()	where likes > 50
Greater Than Equals	{<key>:{$gte:<value>}}	db.mycollection.find({"likes":{$gte:50}}).pr etty()	where likes >= 50
Not Equals	{<key>:{$ne:<value>}}	db.mycollection.find({"likes":{$ne:50}}).pre tty()	where likes != 50

AND in MongoDB

Syntax

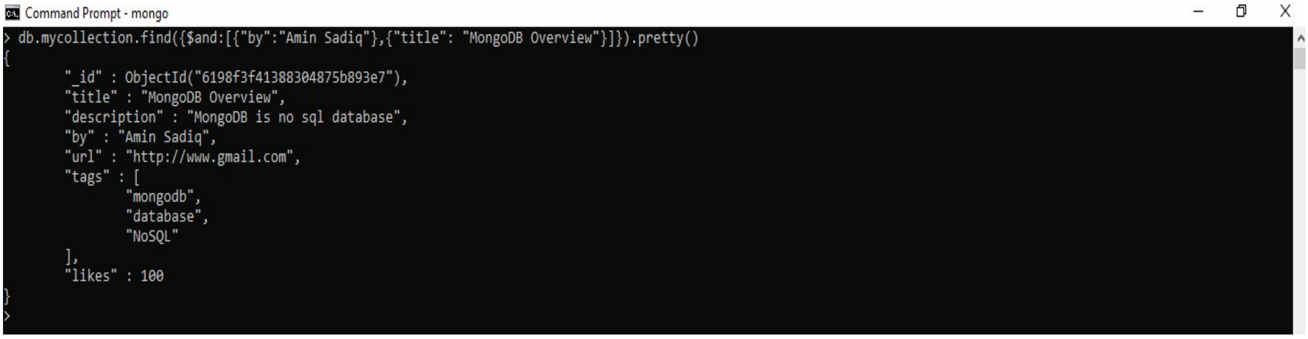
In the find() method, if you pass multiple keys by separating them by ',' then MongoDB treatsit as AND condition. Following is the basic syntax of AND –

```
db.mycol.find(
  {
    $and: [
      {key1: value1}, {key2:value2}
    ]
  }
).pretty()
```

Example

Following example will show all the tutorials written by 'Amin Sadiq' and whose title is 'MongoDB Overview'.

db.mycollection.find({$and:[{"by":"Amin Sadiq"},{"title": "MongoDB Overview"}]}).pretty()



OR in MongoDB

Syntax

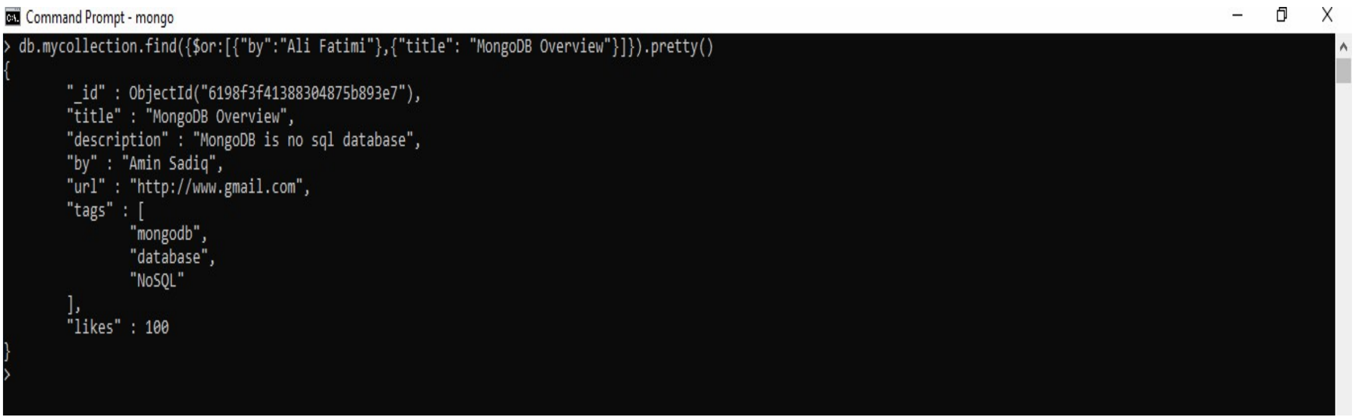
- To query documents based on the OR condition, you need to use $or keyword. Followingis the basic syntax of OR –

```
db.mycol.find(
  {
    $or: [
      {key1: value1}, {key2:value2}
    ]
  }
).pretty()
```

Example

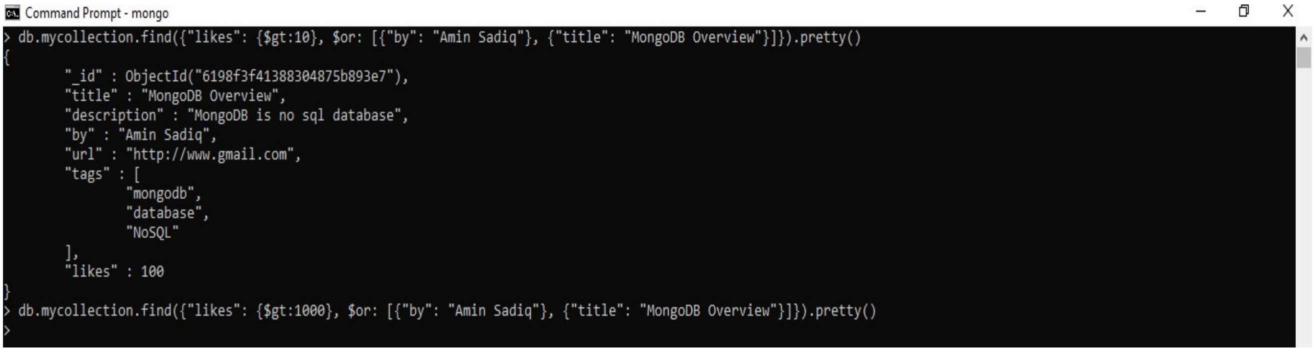
Following example will show all the tutorials written by 'Amin Sadiq' or whose title is 'MongoDB Overview'.

db.mycollection.find({$or:[{"by":"Ali Fatimi"},{"title": "MongoDB Overview"}]}).pretty()



Using AND & OR together in MongoDB

The following example will show the documents that have likes greater than 10 and whose title is either 'MongoDB Overview' or by is 'Amin Sadiq'. Equivalent SQL where clause is 'wherelikes>10 AND (by = 'Amin Sadiq' OR title = 'MongoDB Overview')' db.mycollection.find({"likes": {$gt:10}, $or: [{"by": "Amin Sadiq"}, {"title": "MongoDB Overview"}]}).pretty()



MongoDB – Update Document

MongoDB Update() Method

The update() method updates the values in the existing document

Syntax

- The basic syntax of update() method is as follows –

db.COLLECTION_NAME.update(SELECTION_CRITERIA, UPDATED_DATA)

Example
Consider the mycollection collection has the following data.

{ "_id" : ObjectId(5983548781331adf45ec5), "title":"MongoDB Overview"}
{ "_id" : ObjectId(5983548781331adf45ec6), "title":"NoSQL Overview"}
{ "_id" : ObjectId(5983548781331adf45ec7), "title":"Tutorials Point Overview"}

Following example will set the new title 'New MongoDB Tutorial' of the documents whose titleis 'MongoDB Overview'.

db.mycollection.update({'title':'MongoDB Overview'},{$set:{'title':'New MongoDB Tutorial'}})

Command Prompt - mongo

```
> db.mycollection.update({'title':'MongoDB Overview'},{$set:{'title':'New MongoDB Tutorial'}})
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
> db.mycollection.find().pretty()
{
        "_id" : ObjectId("6198f3f41388304875b893e7"),
        "title" : "New MongoDB Tutorial",
        "description" : "MongoDB is no sql database",
        "by" : "Amin Sadiq",
        "url" : "http://www.gmail.com",
        "tags" : [
                "mongodb",
                "database",
                "NoSQL"
        ],
        "likes" : 100
}
{
        "_id" : ObjectId("6198f3f41388304875b893e8"),
        "title" : "NoSQL Database",
        "description" : "NoSQL database doesn't have tables",
        "by" : "Ali Shah Fatmi",
        "url" : "http://www.gmail.com",
        "tags" : [
                "mongodb",
                "database",
                "NoSQL"
        ],
        "likes" : 20,
        "comments" : [
                {
                        "user" : "user1",
                        "message" : "My first comment",
                        "dateCreated" : ISODate("2021-12-09T21:35:00Z"),
                        "like" : 0
                }
        ]
}
>
```

MongoDB – Delete Document

MongoDB remove() Method

- MongoDB's remove() method is used to remove a document from the collection. remove() method accepts two parameters. One is deletion criteria and second is justOneflag.
- deletion criteria – (Optional) deletion criteria according to documents will be removed.
- justOne – (Optional) if set to true or 1, then remove only one document.

Syntax

Basic syntax of remove() method is as follows –

db.COLLECTION_NAME.remove(DELETION_CRITERIA)

Example

Let's consider the mycollection collection has the following data.

{ "_id" : ObjectId(5983548781331adf45ec5), "title":"New MongoDB Tutorial"}
{ "_id" : ObjectId(5983548781331adf45ec6), "title":"NoSQL Overview"}
{ "_id" : ObjectId(5983548781331adf45ec7), "title":"Tutorials Point Overview"}

Following example will remove all the documents whose title is 'MongoDB Overview'.

db.mycollection.remove({'title':'New MongoDB Tutorial'})

Command Prompt - mongo

```
> db.mycollection.remove({'title':'New MongoDB Tutorial'})
WriteResult({ "nRemoved" : 1 })
> db.mycollection.find().pretty()
{
        "_id" : ObjectId("6198f3f41388304875b893e8"),
        "title" : "NoSQL Database",
        "description" : "NoSQL database doesn't have tables",
        "by" : "Ali Shah Fatmi",
        "url" : "http://www.gmail.com",
        "tags" : [
                "mongodb",
                "database",
                "NoSQL"
        ],
        "likes" : 20,
        "comments" : [
                {
                        "user" : "user1",
                        "message" : "My first comment",
                        "dateCreated" : ISODate("2021-12-09T21:35:00Z"),
                        "like" : 0
                }
        ]
}
>
```